

A highly accurate and interpretable gradient boosting machine learning model for predicting permeability in strongly heterogeneous reservoirs using the Optuna framework

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Abstract

Permeability, a key parameter for evaluating the capability of fluid flow within reservoir pores, significantly influences the calculation of effective reservoir thickness, reserves assessment, and production evaluation. While logging techniques offer detailed characterization of the physical properties of reservoir rocks, constructing an accurate permeability prediction model remains a technical challenge. Conventional logging methods can predict permeability for homogeneous layers but face significant limitations when dealing with more heterogeneous layers. Therefore, a novel hybrid method combining Optuna and a light gradient boosting machine (LightGBM) has been proposed for predicting permeability in heterogeneous reservoirs. During the prediction process, Spearman's rank correlation coefficient was used to select 11 well logging curves from a strongly heterogeneous sandstone reservoir as training features. Permeability was used as the target variable for training. The effectiveness of the trained model was evaluated using the root mean square error (RMSE), mean absolute error (MAE), and explained variance score (EVS). The RMSE, MAE, and EVS of the train set were 0.0158, 0.0631, and 0.9863, respectively. The RMSE and MAE were similar, and EVS was 0.8698 of the test set. To further verify the model's reliability, permeability predictions were made using blind well data from the area, and the results were analyzed. The analysis showed that the predicted permeability values closely matched the actual core permeability measurements. Additionally, the Shapley additive explanations method was used to interpret the model, and the resulting explanations were consistent with geological understanding. This indicates that the Optuna-LightGBM model possesses high precision and good applicability in the area.

Keywords: machine learning; permeability prediction; Optuna framework; LightGBM; Shapley additive explanations

1. Introduction

Permeability is one of the most fundamental and critical parameters for measuring reservoir characteristics, as it directly influences the efficiency of fluid transmission and the storage capacity within the reservoir. It plays an important role in assessing reservoir productivity and calculating effective thickness, among other aspects (Wu *et al.* 2023). In the petroleum and natural gas industry, core sampling analysis is commonly used to determine the reservoir's permeability value. However, direct measurement methods are not only costly and time consuming but also limited to specific reservoir sections (Sun *et al.* 2021). Theoretical analysis can enhance the understanding of the microscopic properties of rocks (Chen *et al.* 2023; Cai *et al.* 2024; Dou *et al.* 2024; Chen *et al.* 2025), but it needs to be combined with practical applications in oil and gas fields. Therefore, developing a method that is rapid, economical, and precise for obtaining permeability values has become a critical issue in the field of petroleum exploration and development.

Well logging data, as a continuous and uniform source of high-resolution information, provides detailed insights into subsurface rock properties. Establishing a correlation model between reservoir permeability and well logging data is essential for accurately predicting permeability. The main methods currently used to derive permeability from well logging data include empirical formula methods, nuclear magnetic resonance (NMR)-based models, and machine learning methods (Latief & Fauzi 2012; Ge *et al.* 2014; Trevizan *et al.* 2014; Sudakov *et al.* 2019; Subasi *et al.* 2022; Matinkia *et al.* 2023; Stefanelli *et al.* 2023; Haritha *et al.* 2025). The empirical formula methods rely on extensive analysis and induction of logging data, such as fitting porosity-permeability data, which provide a foundation for the quantitative interpretation of logging data. There are alternative empirical formulas available, such as those proposed by Ge *et al.* (2014), which are based on the Rose and Al-Fattah empirical formulas and can be applied to permeability calculations in low-permeability reservoirs. Latief & Fauzi (2012) proposed a new exponent law

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for porosity, offering more accurate permeability predictions for most models. However, empirical formulas may have limited accuracy and applicability in complex reservoirs due to their reliance on pore structure. NMR-based models, on the other hand, can effectively differentiate between free water and bound water, providing key parameters for permeability calculations, making it a valuable tool for permeability prediction research. Trevizan *et al.* (2014) used core permeability data and NMR logging measurements to construct a permeability prediction model based on radial basis function, achieving good results in predicting the permeability of complex carbonate reservoirs. Stefanelli *et al.* (2023) employed NMR logging and core analysis techniques to propose a novel method for estimating pore size distribution and the Timur-Coates permeability curve based on a T_2 spectrum and laboratory measurement data. Despite the high precision of permeability values provided by NMR-based models, they are costly and time consuming, requiring complex data processing and interpretation.

With the rapid and advancement of artificial intelligence technology, research on integrating well logging with machine learning is becoming increasingly popular (Ahmadi & Chen 2019; Hu *et al.* 2023; Li *et al.* 2024; Rahmati *et al.* 2024; Djamel *et al.* 2025). Machine learning can be used to predict permeability. Al-Anazi & Gates (2010) and Gholami *et al.* (2012) utilized support vector regression for permeability prediction, marking one of the early applications of machine learning in this field. Despite the enhanced capabilities of these methods, their efficiency remains low, necessitating the search for alternative methods for permeability prediction. As machine learning models have evolved, Adeniran *et al.* (2019) proposed an integrated learning method combining K -nearest neighbor and nearest cluster center mechanisms to predict permeability. Their study compared heterogeneous and homogeneous integrated models and confirmed that the heterogeneous integrated model is more effective in permeability prediction. Zhang & Cai (2021) developed a model using digital image analysis and rock type methods, combined with an indirect prediction method employing backpropagation neural networks and random forest algorithms to more accurately predict permeability. Although support vector regression and random forest algorithms have powerful capabilities and can effectively address issues related to small sample sizes and overfitting, they have significant limitations in parameter selection and handling low-porosity and low-permeability reservoirs (Al-Anazi & Gates 2010; Gholami *et al.* 2012; Han *et al.* 2021). To address this issue, Gu *et al.* (2021) proposed a combination of particle swarm optimization and the XGBoost machine learning model (PSO-XGBoost) to predict the permeability of complex geological reservoirs, thereby improving the accuracy and efficiency of permeability prediction. Subasi *et al.* (2022) developed a stochastic gradient boosting regression model to predict the permeability of oil reservoirs using NMR logging data, achieving high accuracy and efficiency. Zhao *et al.* (2022) employed an XGBoost machine learning model in conjunction with logging data to effectively and rapidly predict the permeability of low-permeability sandstone without reliance on core test data. Among the factors, acoustic travel time and corrected compensated neutron logging parameters were found to be the most contributive, offering a new approach for underground resource assessment. However, the gradient boosting machine learning methods mentioned are not suitable for highly heterogeneous

reservoir sections. Most of these methods focus primarily on enhancing model accuracy with less emphasis on model interpretability, which may limit the practical application of machine learning methods in actual permeability prediction.

To achieve highly accurate and interpretable capabilities, we developed a novel hybrid machine learning method combining Optuna automatic hyperparameter optimization framework and gradient boosting algorithm light gradient boosting machine (LightGBM) for predicting permeability in heterogeneous reservoirs. First, by using Optuna to automatically identify the optimal hyperparameter combination, these hyperparameters were applied to LightGBM, resulting in the construction of the Optuna-LightGBM model. Second, based on the well logging data from a low-permeability, strongly heterogeneous sandstone reservoir, the Optuna-LightGBM model was compared with three tree-based boosting algorithms: gradient boosting decision tree (GBDT), extreme gradient boosting (XGBoost), and LightGBM. Finally, to address the challenge of transparency in machine learning, we employed the SHAP model to analyze the local and global explainability of the trained Optuna-LightGBM model.

2. Methods of Optuna-LightGBM framework

2.1. Automatic hyperparameter optimization software framework (Optuna)

Optuna is an open-source software framework designed specifically for automatic hyperparameter optimization. Proposed by Akiba *et al.* (2019), it consists of several key steps: defining the search space, evaluating the objective function, employing sampling strategies, and continuous feedback optimization. These steps collectively enable the framework to search for the optimal hyperparameter combination through trial and error. Optuna is known for its stability and robust visualization capabilities, allowing users to clearly observe the model's parameter selection and training process. It follows a rigorous iterative optimization process, which can be described in the following five key steps.

Step 1: defining the surrogate model for the objective function. Initially, a probabilistic surrogate model is established to approximate the behavior of the true objective function. This surrogate model estimates the expected performance by exploring the hyperparameter space of the objective function. A commonly used probabilistic model is the Gaussian process (GP) constructed function distribution model (Somla & Bartlett 2000). If a function follows a Gaussian distribution, it is determined by the mean function M and the covariance function k , shown as

$$f(\mathbf{x}) \sim N(M(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \quad (1)$$

where the arbitrary $\mathbf{x}$ corresponds to the probability density function $f(\mathbf{x})$ which follows a normal distribution. The kernel function $k(\mathbf{x}, \mathbf{x}')$ is a key factor in fitting the black-box function $f(\mathbf{x})$ in the GP. Assuming $M(\mathbf{x}) = 0$, we uniformly select several points $X_{1:t}$ and input them into $f(\mathbf{x})$:

$$\begin{bmatrix} \psi(X_{1:t}) \\ \psi(X_{t+1}) \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} k(x_{1:t}, x_{1:t}) & k(x_{1:t}, x_{t+1}) \\ k(x_{t+1}, x_{1:t}) & k(x_{t+1}, x_{t+1}) \end{bmatrix} \right), \quad (2)$$

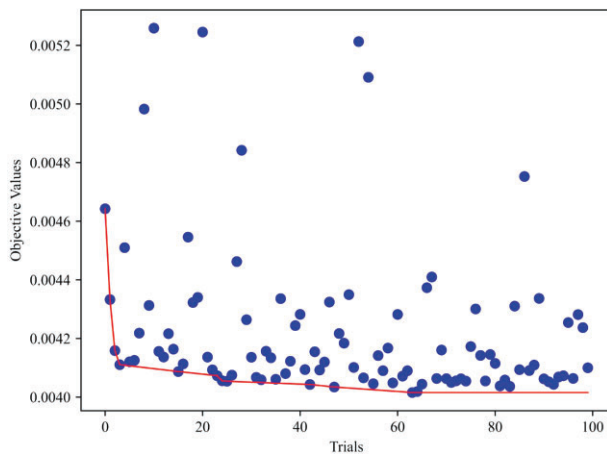


Figure 1. Optimization process history record.

where $\psi(X_{1:t})$ represents the corresponding output that follows a multivariate Gaussian distribution with a mean of 0 and a covariance matrix of $k(x_{1:t}, x_{t+1})$. x_{t+1} is the new sampling point.

Step 2: optimal parameter search on the surrogate model. Based on the constructed surrogate model, an efficient search strategy is adopted to find the potential optimal hyperparameter combination that exhibits the best or near-best performance indicators on the surrogate model.

Step 3: verification and application of target functions. The optimal hyperparameter set identified by the surrogate model is applied to the actual machine learning or model training task target function to evaluate the model performance in reality. In this study, RMSE is used as the target function, and the optimal hyperparameter set is sought in the direction of minimizing RMSE, as shown in Fig. 1. The vertical axis represents the target function (i.e. RMSE), and as the number of training iterations (trials) increases, the target function moves toward a smaller value.

Step 4: online update and pruning strategy for the surrogate model. The surrogate model is updated in real-time based on the actual results of each experiment, to more accurately reflect the true behavior of the target function.

Step 5: Iterative optimization and termination condition. Steps (2)–(4) are iteratively repeated until the maximum op-

timization iteration number is reached. At the end of the iteration, the optimal hyperparameter combination and its corresponding real target function value are retained as the final result.

Bayesian optimization with GPs ensures theoretical convergence in the process of finding the optimal objective function (Shahriari et al. 2016). Optuna distinguishes itself with a powerful search algorithm and flexible hyperparameter space definitions, along with excellent stability and efficient parallel search capabilities, making it a popular choice.

2.2. Gradient boosting method

Gradient boosting is an iterative machine learning technique dominated by three main algorithms: GBDT, XGBoost, and LightGBM. GBDT laid the foundation for optimizing residuals, while XGBoost introduced regularization and parallel computing. LightGBM further advanced this field by achieving breakthroughs in memory usage and training speed through techniques. These advancements have enhanced the practicality and efficiency of the series of algorithms. Therefore, in our research, we integrate LightGBM with Optuna for the prediction of permeability.

LightGBM, introduced by Ke et al. (2017), improves on the XGBoost algorithm by solving issues such as high memory consumption, slow training speeds, and limitations in cache affinity optimization. There are five main principles. (i) A histogram-based decision tree algorithm: as shown in Fig. 2, LightGBM approximates the feature distribution by discretizing features and constructing a histogram, rather than traversing all possible split points of all features. This approach significantly reduces computational load, especially in big data and high-dimensional cases, and decreases memory consumption while improving training speed. (ii) Leaf node growth strategy: traditional GBDT usually grows decision trees using a depth-first or breadth-first approach. However, LightGBM uses the optimal leaf node strategy, where each time the most promising leaf node that can reduce the loss function is selected for splitting. This strategy can find a better solution faster, but may also lead to overfitting, so a depth limit is added to avoid the overfitting problem. (iii) Gradient sampling: to reduce data size efficiently, LightGBM introduced the gradient-based one-side sampling method. This technique selects data samples based on their gradient magnitude, prioritizing instances with larger gradients. By focusing on these informative

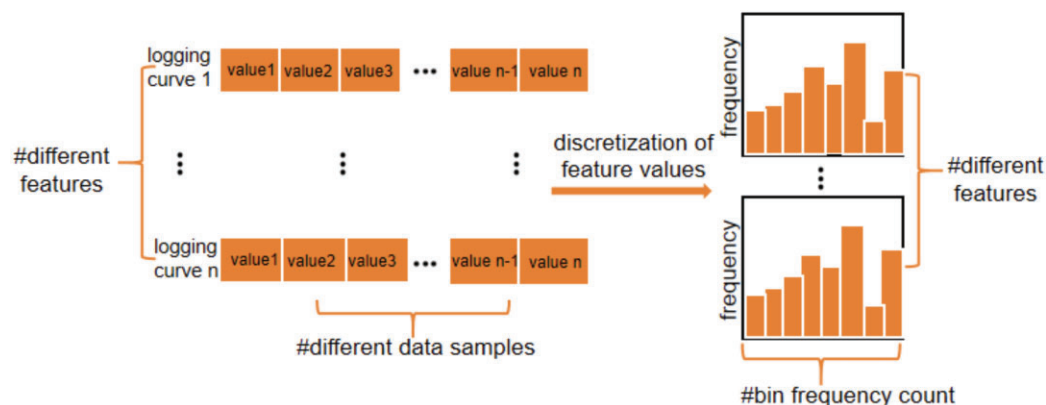


Figure 2. Histogram algorithm.

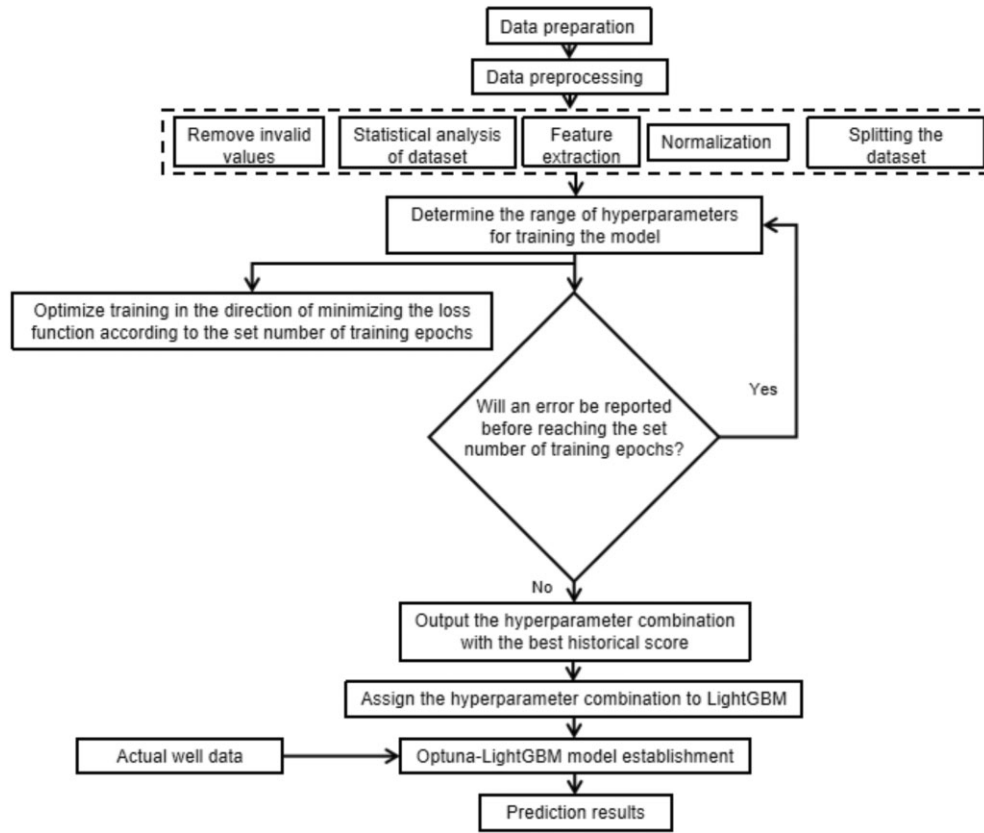


Figure 3. Integration process flow of Optuna and LightGBM.

samples, gradient-based one-side sampling maintains model accuracy while notably reducing the number of training samples needed. (iv) Exclusive feature bundling: when dealing with sparse data, LightGBM detects exclusive features to reduce the feature dimension, effectively compressing the data size and reducing the computational load, especially suitable for large-scale sparse data. (v) Cache optimization and multi-thread computing: LightGBM optimizes cache hit rates during data loading and computation, utilizing modern hardware capabilities to their fullest extent. Additionally, it supports multi-threaded parallel computing, enhancing the efficiency of the training process by effectively utilizing hardware resources.

2.3. Model evaluation indicators

In the field of machine learning, evaluation metrics serve as a measure of model performance, a basis for model optimization and selection, and a guide for model optimization and hyperparameter tuning. Since this study falls under the category of regression problems in machine learning, we use explained variance score (EVS), mean absolute error (MAE), and root mean squared error (RMSE) as evaluation metrics to compare and select models.

- (i) EVS: this measures the percentage of variance in the dependent variable that is explained by the model, with a score ranging from 0 to 1. The closer EVS is to 1, the stronger the model's ability to explain the variation in the dependent variable, and the better the model fits the data. Conversely, a lower EVS indicates

a weaker model. The equation for calculating EVS is as

$$EVS = 1 - \frac{Var\{y_i - \hat{y}_i\}}{Var\{y_i\}}, \quad (3)$$

where Var represents the variance, y_i is the actual value of the i th sample, and $\hat{y}_i$ is the predicted value of the i th sample.

- (ii) MAE: the average of the absolute values of the prediction errors, indicating the average degree of deviation between the predicted results and the actual results. The calculation formula is

$$MAE = \frac{1}{m} \sum_{i=1}^m |\hat{y}_i - y_i|. \quad (4)$$

- (iii) RMSE: one of the commonly used metrics to evaluate the performance of a machine learning model. It measures the square root of the average of the squared differences between the model's predicted values and the actual observed values, and can be considered the square root of the Euclidean norm (L_2 norm) of the prediction error. The calculation formula is shown in the following:

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2}. \quad (5)$$

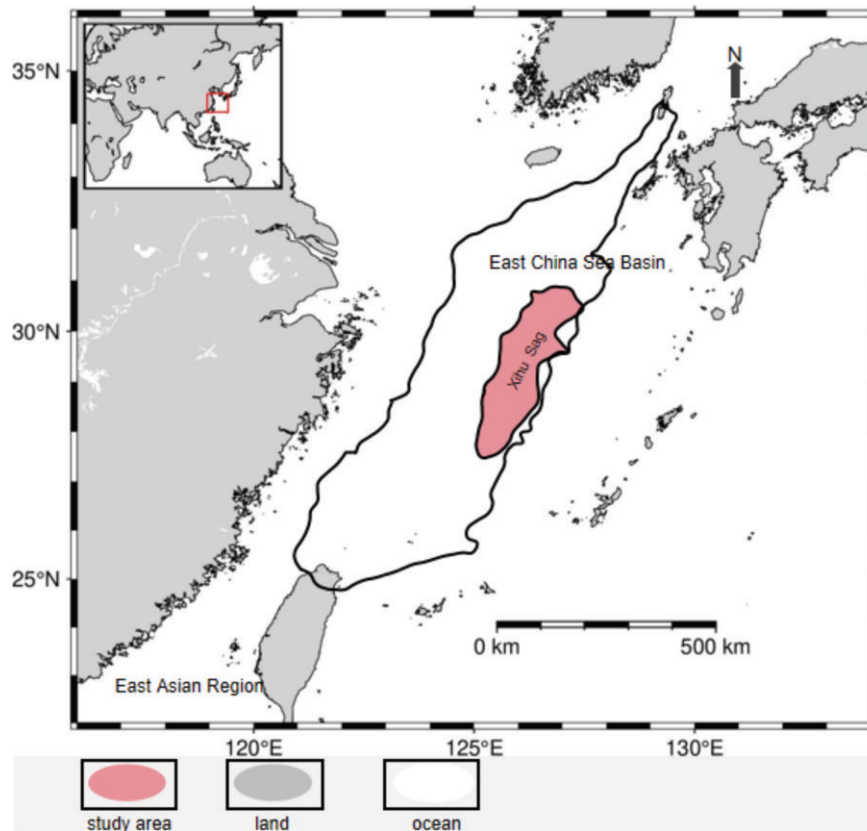


Figure 4. Location map of the Xihu Sag in the East China Sea Basin.

2.4. Automatic hyperparameter optimization with Optuna and LightGBM integration

In Sections 2.1 and 2.2, we introduced the methods of the hyperparameter optimization framework Optuna and the LightGBM algorithm, respectively. Their integration is illustrated in Fig. 3. The integration of Optuna with LightGBM begins with data preparation, followed by comprehensive preprocessing steps. This includes handling invalid values, analyzing distribution characteristics, extracting training features, normalizing data, and partitioning the dataset. Subsequently, well-log curves that show strong correlation with the label (permeability) are selected. The label and feature data are then combined as input for the model, which is integrated into the Optuna framework for training and hyperparameter tuning. Within the Optuna framework, the objective function and its associated loss function are predefined, along with specific ranges for hyperparameter search. After each iteration, the system computes the current loss and compares it with losses from previous iterations. If the Optuna process fails to find an optimal solution within the preset maximum iterations, termed a “trial failure,” it restarts with a new set of iterations. On reaching the maximum predefined training iterations (set to 100 in this study), Optuna records and outputs the historical optimal loss score and its corresponding optimal hyperparameter combination. It is important to note that the maximum training iteration can be adjusted as needed based on actual requirements.

These optimal hyperparameters are then applied to LightGBM, which undergoes additional training iterations. The success of constructing the model using the Optuna-

LightGBM algorithm is determined by achieving an EVS of 0.8, as well as sufficiently low values for RMSE and MAE. Ultimately, the integrated Optuna-LightGBM model is used to predict reservoir permeability.

3. Application results

3.1. Well logging data collection and analysis

The East China Sea Basin, located in the nearshore region off the eastern coast of China, represents a post-rift to sag basin that emerged during the Mesozoic and Cenozoic periods. The tectonic evolution of this basin is intricately linked to the subduction of the Pacific Plate beneath the Eurasian Plate, shown in Fig. 4. Data for this study are derived from the Xihu Sag, where the tight sandstone layers of the Paleogene Pinghu Formation and the Neogene Huagang Formation are recognized as the main targets for hydrocarbon exploration in the area (Zeng *et al.* 2021; Dong *et al.* 2023).

The well logging dataset used in this study is collected from Well #A in the East China Sea Basin Xihu Sag. The well logging curves include spontaneous potential (SP), natural gamma (GR), drill bit diameter (DBD), caliper (CAL), compensated neutron log (CNL), compensated density log (DEN), compressional sonic travel time (DTC), shear sonic travel time (DTS), phase resistivity at 16 inches (P16H), phase resistivity at 28 inches (P28H), phase resistivity at 40 inches (P40H), attenuation resistivity at 40 inches (A40H), shale content (SH), caliper variation (CV), porosity log curve (POR_LOG), and depth (DEPTH). A total of 1767 core plugs are collected from Well #A for laboratory permeability measurements.

Table 1. Well logging data statistics of Well #A.

Curve name	DEPTH (m)	SP (mV)	GR (API)	DBD (in)	CAL (in)	CNL (%)	DEN ($\Omega\cdot\text{m}$)	DTC ($\mu\text{s}/\text{ft}$)	DTS ($\mu\text{s}/\text{ft}$)
log value	4211.44	57.10	45.34	8.5	9.63	4.82	2.49	73.14	117.61
	4211.54	57.03	45.44	8.5	9.63	4.99	2.49	81.26	117.39
	4211.63	56.97	45.68	8.5	9.63	5.22	2.49	83.08	117.17
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	3605.00	56.84	47.37	8.5	9.62	5.66	2.50	75.57	131.91
	3610.00	56.84	47.37	8.5	9.62	5.66	2.50	75.57	131.91
total number	2767	2767	2767	2767	2767	2767	2767	2767	2767
mean	3743.69	-134.27	61.00	10.05	10.27	8.33	2.49	69.20	111.95
variance	277.91	208.20	14.25	1.84	1.73	4.25	0.06	4.91	10.06
minimum	3124.20	-426.72	35.86	1.97	8.17	0.93	2.32	57.86	93.14
upper quartile	3606.95	-354.26	51.50	2.08	8.52	5.04	2.45	65.61	106.02
median	3767.77	-205.35	59.30	2.11	9.57	7.66	2.49	68.55	110.11
lower quartile	3966.95	69.09	67.60	2.18	12.12	10.70	2.53	70.84	114.75
maximum value	4331.00	116.03	176.43	2.39	15.72	32.52	2.69	88.73	149.29
curve name	P16H ($\Omega\cdot\text{m}$)	P28H ($\Omega\cdot\text{m}$)	P40H ($\Omega\cdot\text{m}$)	A40H ($\Omega\cdot\text{m}$)	SH (%)	CV (in)	POR_LOG (%)	PERM (mD)	
log value	9.19	12.45	15.98	27.73	9.18	4.82	8.51	0.18	
	9.24	12.39	15.85	27.82	9.22	4.99	8.58	-0.05	
	9.33	12.26	15.86	27.74	9.33	5.22	8.64	0.04	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
	9.44	12.49	16.37	28.02	10.09	5.66	8.72	0.08	
	9.44	12.49	16.37	28.02	10.09	5.66	8.72	0.28	
total number	2767	2767	2767	2767	2767	2767	2767	2767	
mean	14.85	19.96	24.26	34.14	8.43	0.38	9.69	0.21	
variance	9.80	12.84	14.99	21.35	8.22	0.46	3.32	0.85	
minimum	1.81	2.62	3.86	9.23	0.00	0	0	-1.87	
upper quartile	7.55	10.06	11.78	20.07	3.77	0.11	7.56	-0.44	
median	11.60	16.07	19.78	26.16	6.68	0.23	9.67	0.16	
lower quartile	20.56	27.40	33.60	44.71	10.04	0.44	11.47	0.79	
maximum value	60.01	138.47	74.65	172.07	100.00	3.86	17.88	2.93	

The statistical characteristics of the original data are shown in Table 1, which lists the statistical features of the dataset, including the maximum, minimum, median, variance, mean, upper quartile, and lower quartile values. The study area is a low-permeability, strongly heterogeneous sandstone reservoir, affected by factors such as lithology and pore structure. Conventional permeability calculation methods are not applicable to this well. Therefore, we apply different gradient boosting machine learning models to predict permeability using the well logging curves mentioned before.

3.2. Data preprocessing and model selection

In machine learning, data preprocessing is the first and important step, which directly affects the accuracy of the subsequent model training. For Well #A, the preprocessing steps mainly include the following procedures.

- (i) Removing invalid values. The dataset from Well #A is primarily derived from formations characterized by sandstone and mudstone. During drilling operations, geological complexities, and uncertainties resulted in severe borehole enlargement issues in specific intervals of the formation. This has significantly distorted the well logging curves, making them unsuitable for accurate and comprehensive evaluation of the reservoir properties. To ensure data quality, we employ a

method that involves identifying and removing data from expanded sections, denoted CV:

$$CV = |CAL - DBD|, \quad (6)$$

where CAL represents the wellbore curve, DBD represents the bit diameter, and CV indicates the hole enlargement, with each measured in inches.

- (ii) Statistical analysis and data processing. After removing invalid values during the preprocessing step, the statistical analysis of the dataset is shown in Fig. 5. Here, the horizontal axis represents the logging curves and their respective units, while the vertical axis shows the frequency distribution. We see that the data distribution characteristics of each logging curve are different. The DBD data is distributed in two different values. Due to the wide distribution range and multi-order-of-magnitude difference of the permeability, we use the logarithm form of permeability as the object of model learning and prediction to achieve a more precise and quantitative treatment.
- (iii) Extraction of features from the dataset. While well logging curves contain geological information and play an important role in predicting permeability using machine learning methods, including all these

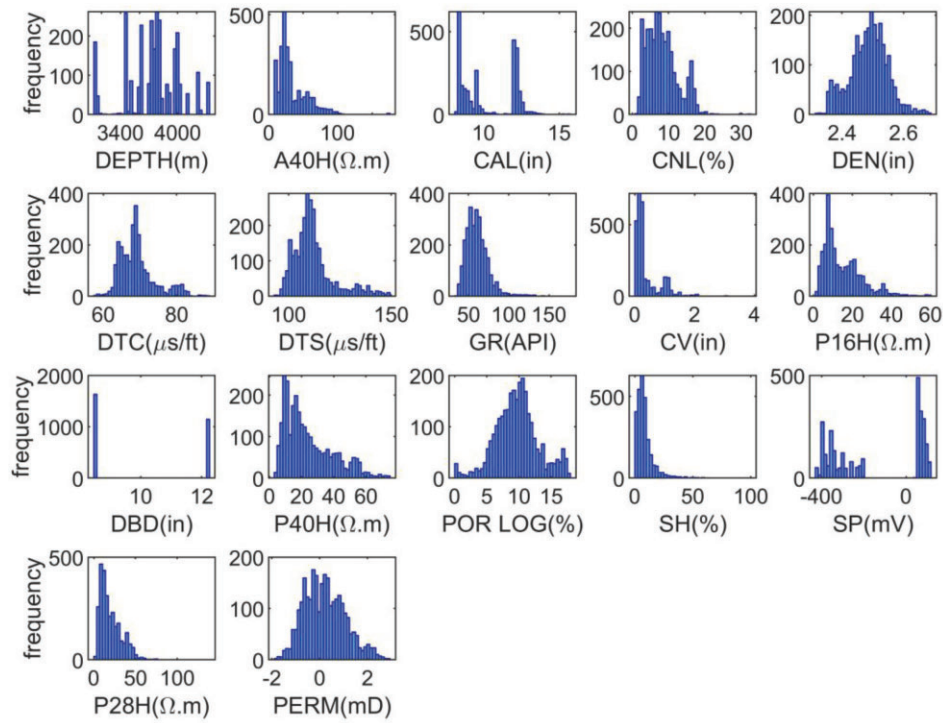


Figure 5. Statistical analysis of the Well #A data set.

curves as inputs can lead to the curse of dimensionality and potentially decrease result accuracy. As shown in Fig. 5, the well logging curves do not conform to a normal distribution; therefore, Spearman's rank correlation coefficient is employed to select features, as shown in Fig. 6. Darker colors in the visualization indicate stronger correlations. Figure 6 illustrates that the correlation coefficients between variables SP, DBD, CAL, A40H, and PERM are each <0.3 , suggesting a weak relationship with permeability. Therefore, these variables are candidates for removal from further analysis. Further analysis reveals significant collinearity between P16H and P28H, as well as between P28H and P40H, with correlations exceeding 0.95. Consequently, P28H is excluded from the analysis. Additionally, the correlation coefficients of DEN and POR_LOG with the permeability indicator are 0.73 and 0.74, respectively. This indicates a relatively strong correlation between them and permeability. It can be inferred that DEN and POR_LOG have a relatively strong explanatory power and make independent contributions to permeability. Therefore, they are incorporated into the model construction process. As a result, the selected features for training include DEPTH, GR, CNL, DEN, DTC, DTS, P16H, P40H, SH, CV, and POR_LOG, with PERM as the target variable.

- (iv) Dataset partitioning. Following the successful acquisition and preprocessing of the original well logging data, it is common practice to partition the dataset into two distinct subsets: the training set and the test set. The training set is used to estimate parameters and construct models within machine learning algorithms, while the trained models are validated for

performance and assessed for generalization capabilities on the test set. Typically, a 7:3 ratio is commonly used to split the dataset into training and test sets, employing random sampling techniques for effective partitioning. However, due to significant variations in sampling density across different orders of magnitude for permeability, traditional random sampling methods may lead to underrepresentation of intervals with low data volume. This could directly affect the distribution characteristics between the training and test sets, potentially compromising the validity and accuracy of the model training process and ultimately making it difficult to achieve the desired outcomes.

Therefore, in this study, permeability data are categorized into five intervals based on different orders of magnitude: 0.01–0.1 mD, 0.1–1 mD, 1–10 mD, 10–100 mD, and 100–1000 mD, as illustrated in Fig. 7. The numbers within each interval correspond to the total number of samples in that range. Data partitioning and sampling are conducted within each permeability interval following a 7:3 ratio.

- (v) Normalization. The data are normalized according to Equation (7) to ensure that all indicators are on the same order of magnitude and to mitigate the adverse effects caused by anomalous samples:

$$X_j = \frac{X - X_{\min}}{X_{\max} - X_{\min}}, \quad (7)$$

where X_j represents the result of the normalized parameter; $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of X , respectively.

By following these five steps, data preprocessing can be completed. Subsequently, the GBDT, XGBoost, LightGBM,

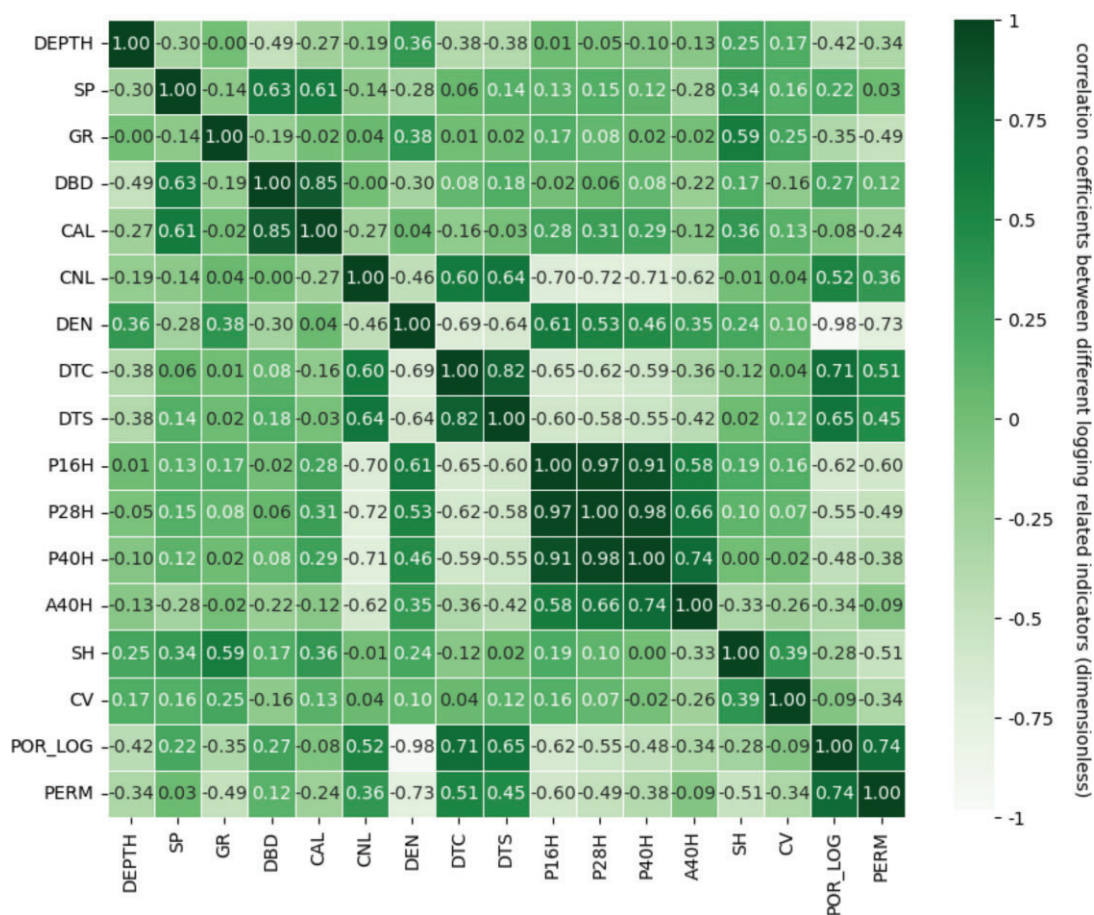


Figure 6. Spearman correlation coefficient matrix.

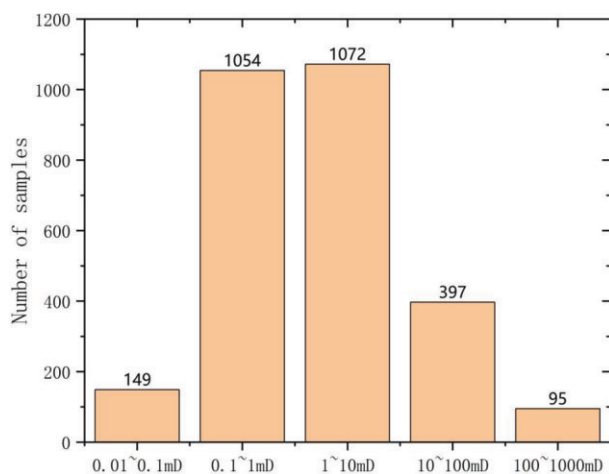


Figure 7. Permeability range division and statistics.

and Optuna-LightGBM regression models are applied to the well logging data from Xihu Sag of the East China Sea Basin. The model training results are analyzed using three evaluation metrics: EVS, MAE, and RMSE, as shown in Table 2. It is evident that for both the training and testing sets, the RMSE and MAE results of the Optuna-LightGBM model after training fall between 0.01 and 0.05, making them the smallest among the four models. Regarding the EVS metric, the Optuna-LightGBM model also shows the highest values for

both the training and testing sets. The EVS for the training set is 0.9863, and for the testing set, it is 0.8698, which are close to 1, indicating that the performance of the Optuna-LightGBM model is good.

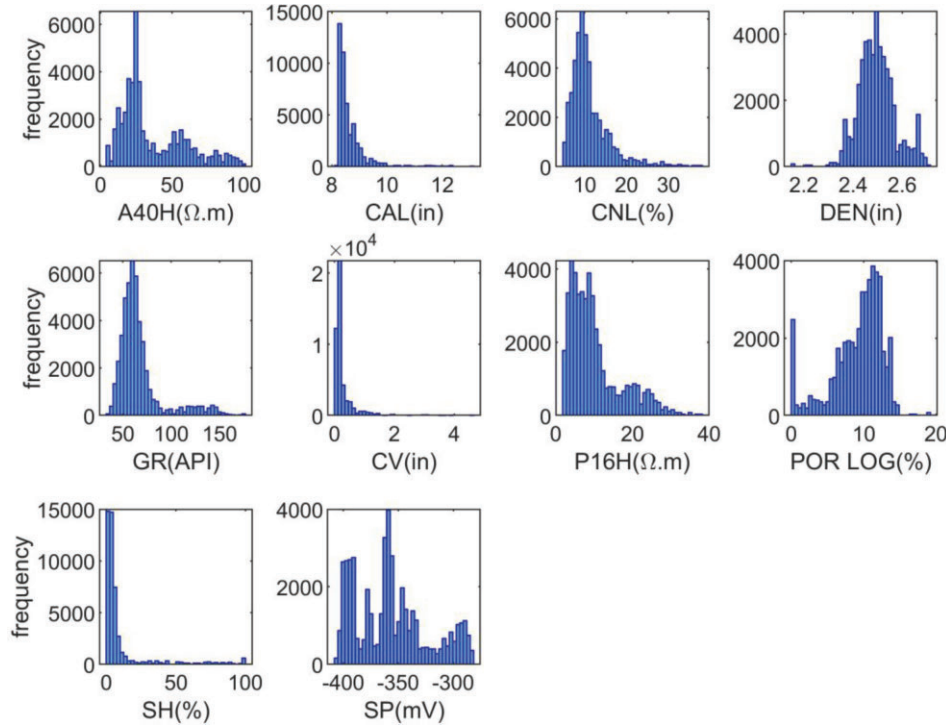
3.3. Prediction results and analysis

In the process of assessing the permeability of tight sandstone reservoirs in our study area, we apply the hybrid Optuna-LightGBM model to Well #B in the Xihu Sag of the East China Sea Basin, as shown in Fig. 8. Figure 8 presents the distribution of well logs in Well B. Compared with the distribution of well logs in Well A, parameters such as GR, CNL, DEN, P16H, A40H, SH, CV, and POR_LOG all show differences. This indicates that the data of Well B are unique. When used to test the prediction of permeability in heterogeneous reservoirs, it can more effectively verify the accuracy and reliability of relevant prediction methods, thus providing strong data support for the study of permeability in heterogeneous reservoirs.

In the process of assessing the permeability of tight sandstone reservoirs in our study area, we apply the hybrid Optuna-LightGBM model to Well #B in the Xihu Sag of the East China Sea Basin, as shown in Fig. 9. To verify and compare the prediction effectiveness, we also apply traditional GBDT, XGBoost, and LightGBM models to the same dataset to predict permeability. By carefully analyzing Fig. 9 (the fifth panel), it is evident that the Optuna-fused and optimized LightGBM model predicts permeability results

Table 2. Evaluation metrics after training with permeability labels.

Regression model	RMSE (training set)	RMSE (test set)	MAE (training set)	MAE (test set)	EVS (training set)	EVS (test set)
GBDT	0.0490	0.0648	0.0336	0.0450	0.9234	0.8662
XGBoost	0.0288	0.0644	0.0199	0.0445	0.9635	0.8667
LightGBM	0.0483	0.0643	0.0334	0.0445	0.9255	0.8661
Optuna-LightGBM	0.0158	0.0631	0.0113	0.0430	0.9863	0.8698

**Figure 8.** Statistical analysis of the Well #B data set.

that better fit the actual laboratory measured permeability data, as indicated by the red circles. Figure 10 illustrates the distribution of predicted permeability values from the four models compared to the actual core permeability values. It is clear that the Optuna-LightGBM model provides the best fit. Furthermore, Table 3 shows the quantitative evaluation metrics, including RMSE, MAE, and EVS, which highlight the performance of the Optuna-LightGBM model. In particular, the Optuna-LightGBM model stands out by achieving a significant decrease in RMSE, reducing it by up to 11.57 units compared to the other three models. This reduction underscores the model's advantage in minimizing prediction errors and enhancing overall prediction accuracy. The GBDT model achieved an EVS of 0.2692, indicating it can explain ~26.92% of the variation, reflecting moderate performance. In contrast, XGBoost has an EVS of 0.1556, revealing its lower ability to explain the data variation and a larger prediction error. The EVS of the LightGBM model is 0.2651, similar to GBDT but slightly lower, indicating that both models performed similarly without optimization. However, the Optuna-LightGBM model demonstrates a significant advantage, with an EVS of 0.6132. This indicates that its ability to explain data variation exceeded that of the other models by nearly a

factor of two, emphasizing the importance of effective hyper-parameter optimization for improving model performance.

3.4. Model interpretation based on SHAP

To address the challenge of transparency posed by black-box models in the field of machine learning, we integrate the Optuna and LightGBM frameworks and further employ the SHAP approach to enhance the interpretability of the permeability prediction model. The main focus of this research is to analyze in detail the specific influence mechanisms of 11 well logging curves on the prediction results of permeability. Based on the cooperative game theory framework proposed by Lundberg & Lee (2017), SHAP systematically assesses the marginal effects of each feature through the implementation of the Shapley value concept, thereby achieving a fine-grained dissection of the model's predictive logic. The weighted sum of all original input model feature combinations is conducted, shown as

$$\phi_i(f) = \sum_{U \subseteq X/\{x_i\}} \frac{|U|!(M - |U| - 1)!}{M!} (f_x(U_{x_i}) - f_x(U)), \quad (9)$$

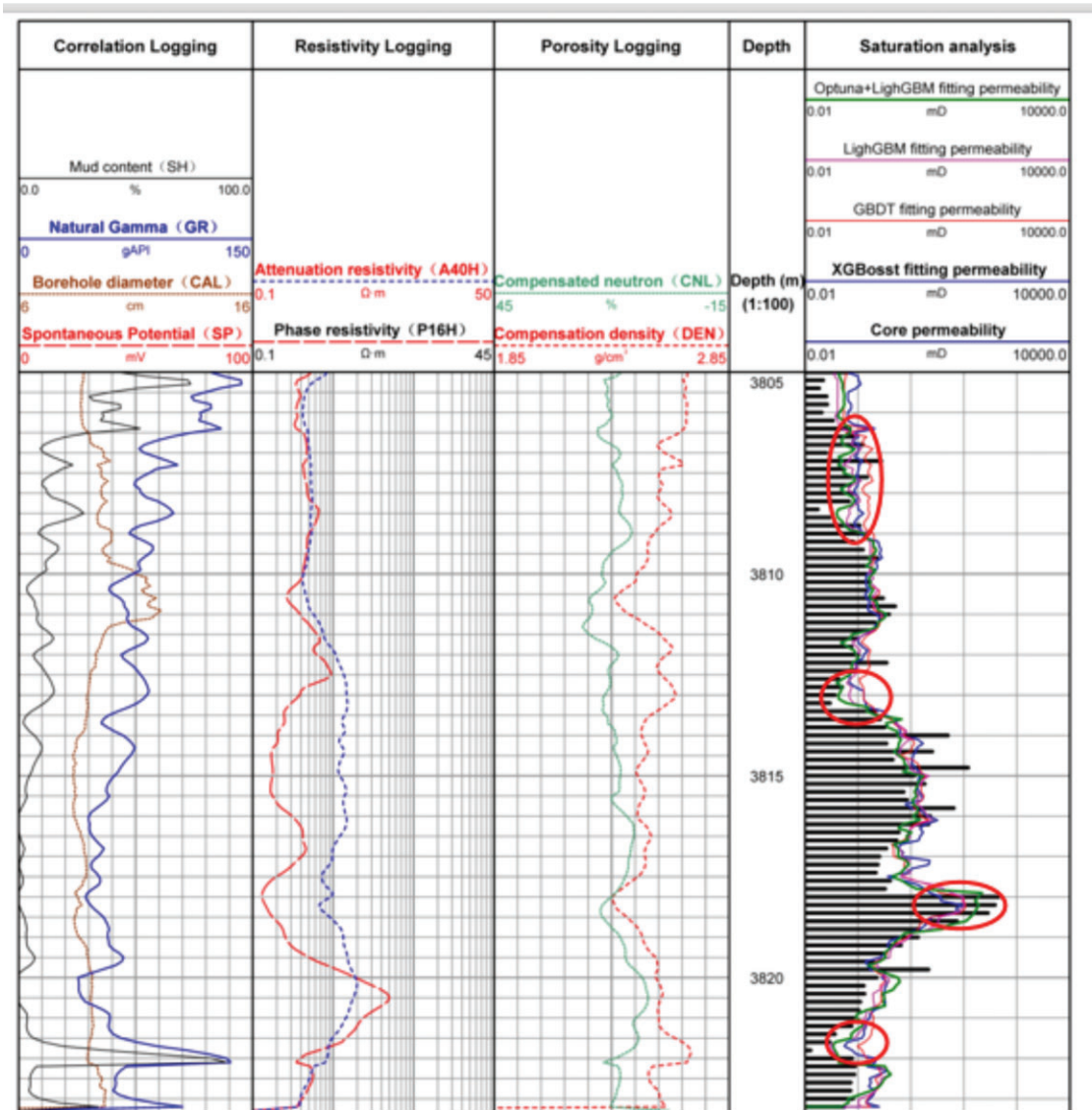


Figure 9. Permeability prediction results from different GBM learning models for Well #B.

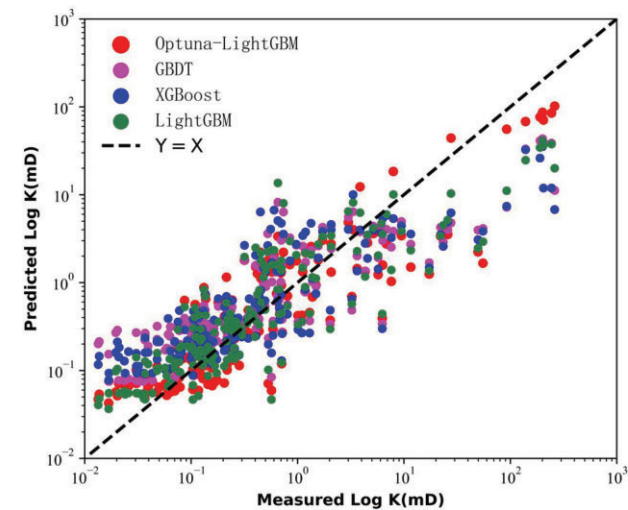


Figure 10. Predicted permeability compared to the measured permeability.

Table 3. Prediction metrics of the four gradient boosting machine learning models.

Regression model	Index		
	RMSE	MAE	EVS
GBDT	33.432	7.850	0.2692
XGBoost	35.899	8.354	0.1556
LighGBM	33.512	7.885	0.2651
Optuna-LighGBM	24.329	6.040	0.6132

where ϕ_i represents the contribution value of the i th feature; f denotes the ensemble model; U is a subset of features in the model; x is the set of all features; x_i is the i th feature; M is the number of features; and $f(U)$ is the prediction for the subset U .

This method conducts a local interpretability analysis of individual prediction values and a global interpretability analysis using feature importance on the integrated

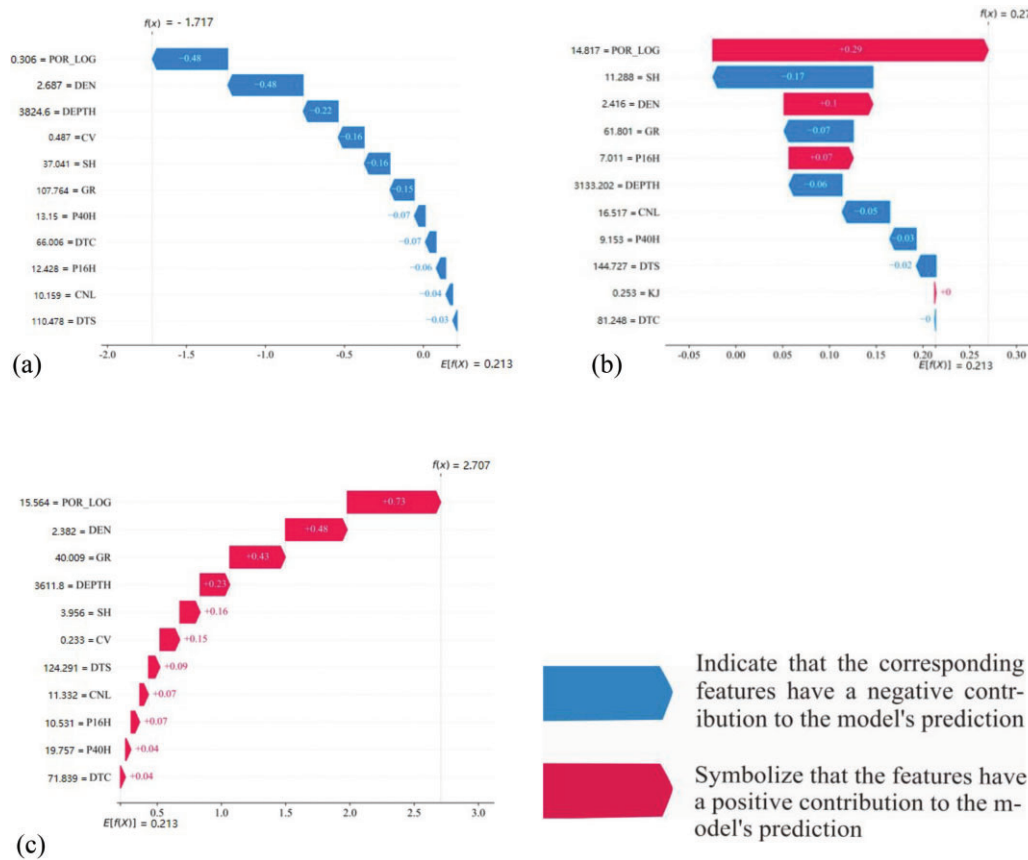


Figure 11. SHAP-based local interpretability-waterfall plot. (a) Prediction minimum value point. (b) Prediction median value point. (c) Prediction maximum value point.

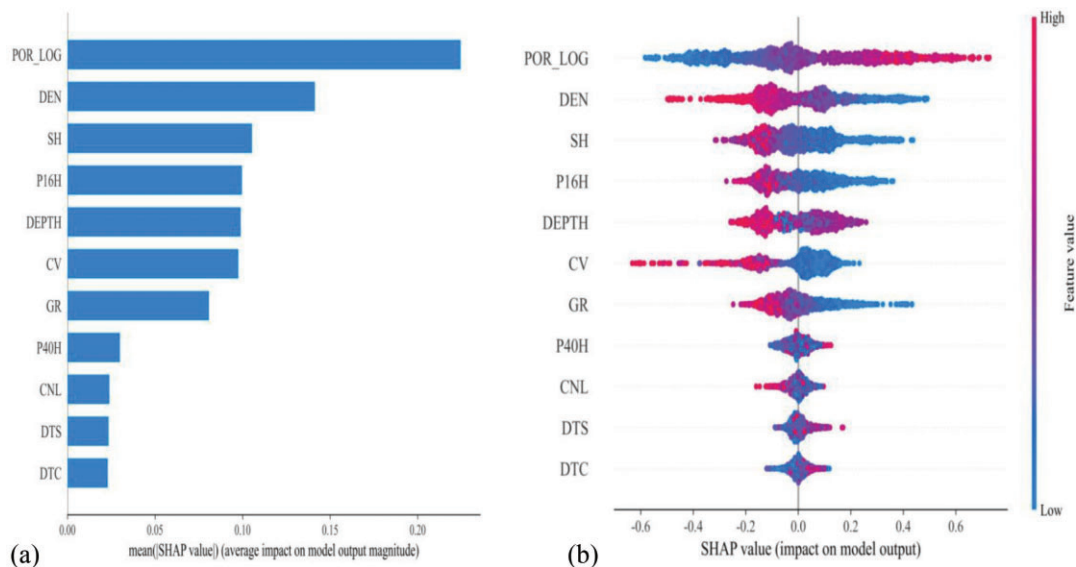


Figure 12. Well logging curve feature importance. (a) Feature importance ranking. (b) Feature summary plot.

Optuna-LightGBM model by using a waterfall plot. In this study, the analysis of the waterfall plot selects the minimum, median, and maximum prediction values of the study area's Optuna-LightGBM model for examination, as shown in Fig. 11. The horizontal axis represents the baseline value of the

interpretative model $E[f(x)]$, which is 0.213; $f(x)$ indicates the output value predicted by the model for a specific sample point, with the specific predictions in Fig. 11a–c being -1.717 , 0.27 , and 2.707 , respectively. The vertical axis displays the feature well logging curves used by the Optuna-LightGBM

model along with their observed actual measurements (annotated in gray). The plot introduces a color-coded arrow mechanism to visually describe the SHAP value contribution of each input feature to the model's final prediction outcome. These color-coded arrows work in concert to reveal how the feature curves, starting from the baseline prediction, push the prediction result to its final state through their respective positive or negative contributions. In Fig. 11, blue arrows indicate a negative contribution of the corresponding feature to the model's prediction, characterized by the arrow pointing to the left, signifying that the feature has reduced the predicted output value; red arrows symbolize a positive contribution from the feature, with the arrow pointing to the right, indicating that these features have increased the model's prediction outcome.

Figure 11 illustrates how each prediction point is influenced by different well logging curves. (i) At the minimum prediction value (Fig. 11a), all well logging features uniformly contribute negatively, resulting in the model predicting values lower than the baseline. (ii) For the median prediction (Fig. 11b), the effects vary: porosity (POR_LOG) makes a significant positive contribution, while shale content (SH) contributes negatively. However, the overall positive contributions dominate, leading to predictions above the model's average level. (iii) At the maximum prediction point (Fig. 11c), almost all well logging features contribute positively, pushing the model's prediction above the baseline value.

Within the SHAP interpretation framework, the absolute magnitude of a feature's SHAP values maps to the strength of its influence. Global feature importance is derived by summarizing the average absolute SHAP values of each feature across the dataset and arranging them in descending order according to this metric, as shown in Fig. 12. This effectively reduces the impact of individual variability. In Fig. 12a, the horizontal axis represents the average feature importance, while the vertical axis lists all well logging curve features included in the model. The length of the blue bars clearly displays the average significance of each feature. The analysis reveals that the most important feature in the Optuna-LightGBM model is the porosity curve POR_LOG, highlighting its dominant contribution to the predicted output, followed by features such as DEN, SH, and P16H. By contrast, the DTC, DTS, and CNL curve features contribute the least to the model's predictive outcome.

While the feature importance plot reveals the ranking of feature influence, it does not show how each feature specifically drives the model's decision-making process and its positive or negative contributions. In contrast, the feature summary plot (Fig. 12b) enhances understanding by displaying the SHAP values of each feature sample, thereby confirming their importance. In this plot, the horizontal axis maps the magnitude of SHAP values, with zero as the central baseline, distinguishing between positive and negative contributions to the model output. Negative values to the left indicate negative contributions, while positive values to the right indicate positive contributions. The vertical axis arranges the well logging curves based on feature importance. Colored dots mark the training samples, with a color gradient (from blue to red) indicating the increase from low to high well logging values. This comprehensive display illustrates the details of how each feature value change positively or negatively influences the model's prediction.

By examining the distribution of sample colors and SHAP values, we can understand how changes in each well logging curve influence the model's predictions. For instance, the key feature, the porosity curve POR_LOG, has a broad distribution that highlights its strong influence on the model: as porosity increases, the corresponding SHAP values also increase, indicating that this feature increases the model's permeability prediction. Second, the density curve DEN shows an inverse relationship, with high DEN values corresponding to low SHAP values, suggesting a decrease in predicted permeability. Similarly, the shale content SH also displays an inverse pattern, with an increase in SH values accompanied by a decrease in SHAP values, diminishing the model's predicted permeability value. These analytical conclusions are consistent with existing geological understanding, strengthening the rationality and reliability of the results. Additionally, for P16H, CV, and GR, a larger value corresponds to a larger SHAP value, indicating a positive contribution to the model's prediction. For DEPTH and CNL, as can be seen from the figure, when their values are smaller, the corresponding SHAP values have both greater than zero and less than zero, indicating that the patterns of DEPTH and CNL are more complex, with uncertain directions of influence on the model.

4. Conclusions

- (i) In this study, we proposed a novel hybrid method combining Optuna and LightGBM for predicting permeability in heterogeneous reservoirs. Four gradient boosting machine learning models were compared, including GBDT, XGBoost, LightGBM, and Optuna-LightGBM. The models are evaluated using three metrics: RMSE, MAE, and EVS. It was found that the Optuna-LightGBM model exhibited the optimal training performance in Well #A for permeability prediction.
- (ii) Using depth, natural gamma, compensated neutron, density, compressional sonic travel time, shear sonic travel time, phase resistivity (P16H and P40H), shale content, hole diameter variation, and porosity curves as features, with permeability as the target variable, four gradient boosting machine learning models were then employed to predict permeability in Well #B. Our proposed model's predictions were more accurate and closely matched laboratory measured permeabilities.
- (iii) We conducted an interpretability analysis of the proposed Optuna-LightGBM model, from local interpretability analysis (SHAP waterfall plot for individual prediction values) to global interpretability (feature importance summary plot). The results were found to be fundamentally consistent with the current geological understanding, which was of significant importance for subsequent model optimization and well logging feature selection.
- (iv) With the advancement of artificial intelligence, the use of machine learning methods for predicting reservoir permeability is becoming increasingly widespread. This research provides valuable reference for future studies aiming to enhance predictive accuracy in reservoir engineering and geological applications.

Author contributions

X.J. was responsible for the investigation, methodology, writing of the original draft, and data analysis. W.Y. was responsible for the funding acquisition, supervision, and review and editing of the draft. H.X. was responsible for acquiring funding, and review and editing of the draft. P.F. was responsible for the methodology, validation, and data analysis. J.S. was responsible for acquiring funding and data analysis. Y.F. helped to write the original draft.

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Conflict of interest statement

None declared.

Data availability

Data will be made available on request.

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